

TO: U.S. DEPARTMENT OF ENERGY (DOE) – OFFICE OF CRITICAL MINERALS AND MATERIALS

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TOPIC: INNOVATION IN U.S. CRITICAL MINERAL SUPPLY CHAINS

INTRODUCTION

Critical minerals such as lithium, cobalt, nickel, rare-earth elements, and gallium are essential to the United States' renewable energy transition, semiconductor manufacturing, and national defense systems.¹ Currently, the United States lists 60 minerals as critical, but domestic production covers only a small portion of this demand. The US is dependent on imports for 12 of these minerals and 50% import-reliant on another 28.² China is dominant in this space, controlling between 50% to 95% of global mineral processing and 69% of rare-earth production.³ This increases supply risks and exposes key U.S. industries to supply disruptions.

Recent innovation across the critical minerals supply chain is expanding new opportunities to address these challenges. Technologies such as AI-enabled mining, deep-sea robotics, advanced mineral processing, and recycling can help expand domestic capacity and reduce long-term dependence on China.

This paper focuses on: How can innovation in critical mineral supply chains strengthen U.S. global competitiveness and reduce dependence on China? It examines the strategic importance of critical minerals, identifies major structural and policy gaps within the U.S. supply chain, assesses emerging technological trends, and proposes policy recommendations aimed at strengthening U.S. critical mineral security and global competitiveness.

BACKGROUND

According to the Energy Act of 2020, critical minerals are materials that are important to the United States' economic and national security, yet face supply chains that can be easily disrupted.⁴ These minerals are important in producing technologies such as EV batteries, defense systems, semiconductors, and renewable energy, and without them, the U.S. would face serious economic and national security risks.⁵ Yet the U.S. still relies heavily on imports for many of these minerals, and most of the processing happens in China.

Examples include minerals essential for clean energy technologies (such as lithium, nickel, cobalt, copper, and graphite), semiconductor manufacturing (including gallium, germanium, cobalt, platinum, palladium, and arsenic), and defense systems (like aluminum, titanium, and tungsten).

This creates a vulnerability as global demand for critical minerals continues to rise due to clean energy, electric vehicles, and digital technologies. Geopolitical tensions, particularly U.S.–China competition and recent supply chain disruptions, have made these risks obvious. While policies such as the Inflation Reduction Act, the CHIPS and Science Act, and the Defense Production Act support domestic supply chains, they remain insufficient to meet long-term demand.

Meanwhile, China continues to dominate the market. It produces most of the world's rare earths, controls over 90% of global refining capacity,⁶ and leads in processing materials such as gallium, graphite, and key battery inputs. Chinese state-backed firms also hold significant stakes in mines worldwide and have previously used this position as leverage in trade disputes. China is even expanding into deep-sea mining, supported by advanced drilling technology and multiple exploration contracts through the International Seabed Authority, an area where the U.S. has limited influence because it has not ratified UNCLOS.⁷ This dominance highlights the urgency for the U.S. to improve its own critical mineral supply chain to protect national security, support clean energy goals, and remain competitive.

PROBLEM STATEMENT: Gaps Threatening U.S. Competitiveness

The main problem is that the U.S. lacks a coordinated innovation strategy across the critical minerals supply chain, which puts it at a competitive disadvantage compared to China. Although advanced technologies for exploration, extraction, processing, and recycling exist, the U.S. has been slow to scale them, and adoption remains fragmented across government, industry, and research institutions. In contrast, China has already integrated these technologies into a coordinated system covering mining, processing, and global investment, giving it long-term strategic control over key materials.

As a result, the United States remains highly dependent on foreign sources, particularly for refining and processing, which weakens its competitiveness in clean energy manufacturing, semiconductors, and defense production. Without faster innovation and stronger domestic capacity, the U.S. risks falling further behind in the global critical minerals market.

CURRENT STATE: INNOVATION IN THE U.S. MINERAL SUPPLY CHAIN

1. AI-Enabled Mineral Exploration and Mining

Artificial Intelligence is already shaping early-stage mineral exploration in the United States by improving accuracy and reducing the time needed to identify viable deposits. U.S. research institutions and private firms now use AI tools to analyze satellite images, geophysical data, and past drilling records, helping geologists identify high-potential areas at lower cost and with less environmental impact. An example is the Department of Energy's National Energy Technology Laboratory AI prospectivity model, which identified a major rare-earth deposit at Wyoming's Brook Mine in under 6 years, showing how AI can shorten discovery timelines.⁸

Automation is also increasingly used in U.S. mining operations. Sites in states such as Nevada now use autonomous haul trucks, remote-controlled drilling, and underground robotics to improve safety, address labor shortages, and maintain steady production.⁹ These systems support efficiency and cost reduction as demand rises for lithium, copper, cobalt, and rare-earth elements. Automation is no longer experimental; it is already part of active U.S. mining operations.

Federal programs are supporting these advances. Projects such as USC's MinMod initiative use AI to combine geological data and improve predictions of undiscovered resources,¹⁰ while the Department of Energy's ARPA-E (Advanced Research Projects Agency-Energy) program funds

AI-enabled recovery of critical minerals from industrial waste and produced water.¹¹ These efforts show early success in expanding domestic supply without relying solely on new large-scale mining. Despite this progress, adoption remains limited compared to global demand and the speed of innovation in competing countries. It is essential to reduce dependence on foreign supply chains and secure domestic manufacturing capacity.

2. Deep-Sea Robotics and Offshore Exploration

Deep-sea robotics and offshore exploration provide the U.S. with another way to access critical minerals beyond land-based sources. Studies show that areas such as the Clarion–Clipperton Zone contain large deposits of nickel, cobalt, manganese, and rare earth elements,¹² which are important for electric vehicles, semiconductors, advanced manufacturing, and defense systems. Because these environments are unsafe for human activity, subsea robotics is central to offshore exploration. Remotely operated vehicles, autonomous underwater systems, and AI-supported seabed imaging now allow accurate mapping, environmental assessment, and sample collection at extreme depths, helping to reduce risk and diversify mineral supply.

Although the United States does not yet have a full deep-sea mining strategy, federal engagement is increasing. Agencies such as the Defense Advanced Research Projects Agency (DARPA), the National Oceanic and Atmospheric Administration (NOAA), and the Office of Naval Research (ONR) research are investing in subsea robotics and ocean-based resource security.¹³ In April 2025, President Trump executive order directed federal agencies to strengthen U.S. leadership in offshore and deep-seabed critical mineral exploration, with emphasis on improved ocean science, streamlined permitting under the Deep Seabed Hard Mineral Resources Act, and domestic processing capacity.¹⁴ Following this, NOAA began a rulemaking process to simplify exploration and recovery permits.¹⁵ While no new licenses have been issued, the order has encouraged firms such as The Metals Company (TMC) to apply for permits under the U.S. framework rather than the international one,¹⁶ despite expected legal and environmental challenges.

Private-sector activity is also increasing. Nauticus Robotics, based in Houston, has secured a \$250 million equity facility to scale its robotics systems for deep-sea mineral exploration.¹⁷ The company plans to use autonomous underwater vehicles and AI navigation tools to conduct surveys and, eventually, support extraction operations¹⁸, and its recent acquisition of SeaTrepid expanded its Remotely Operated Vehicle (ROV) capacity. While U.S. engagement remains at an early stage compared to China, which currently holds multiple International Seabed Authority exploration contracts, the combination of federal support and new private-sector capabilities provides a foundation for further development of offshore mineral resources.

However, to fully benefit from these technologies, the U.S. will need a clearer national strategy, stronger environmental safeguards, and more investment in subsea robotics. These steps are necessary to reduce long-term dependence on foreign mineral supply chains and improve U.S. competitiveness in emerging technology sectors.

3. Advanced Mineral Processing and Recycling

The midstream stage of the critical mineral supply chain, which includes processing, refining, and recycling, remains the United States' weakest point. Although the U.S. has access to important mineral resources, most refining is still carried out in China, which controls up to 90% of global processing for rare earths, lithium, graphite, and several other strategic minerals. This has created a high level of dependence on foreign supply for materials needed in clean energy, defense, and advanced manufacturing. The problem is not about resource availability, but more about the limited number of domestic processing facilities and the high cost of traditional separation technologies. Improving midstream capabilities is therefore essential for reducing import dependence and strengthening the country's competitiveness.

In response, U.S. firms and research institutions are developing cleaner, more efficient, and more flexible processing technologies. These include hydrometallurgical extraction, solvent-free separation, electrochemical refining, and membrane-based filtration. These methods reduce environmental impacts, such as chemical waste and water use, and make it possible to operate smaller, modular processing facilities rather than relying on large, centralized plants. ReElement Technologies, an Indiana-based company supported by federal and Department of Defense funding, is one example. The firm uses advanced chromatography to separate rare-earth oxides in compact facilities and plans to scale its Indiana plant into the largest domestic rare-earth oxide producer by 2026.¹⁹ Although its expected output is still small by global standards, its approach addresses the waste and cost issues that contributed to past U.S. refinery closures, including the shutdown at Mountain Pass.

Recycling technologies are also becoming an important part of building midstream capacity. Closed-loop battery recycling can recover lithium, cobalt, nickel, and graphite at high purity levels, while magnet recycling enables the recovery of neodymium and dysprosium used in electronics, cars, and defense applications. Federal R&D programs are supporting this work. ARPA-E's RECOVER initiatives, for example, are testing bio-based and AI-supported separation methods that extract lithium, cobalt, and rare earths from oilfield brines and mining wastewater.²⁰ These techniques show early success in recovering valuable metals from waste streams that were previously considered too low-grade to process, expanding the range of materials available for domestic supply.

While these efforts demonstrate progress, the United States will need to adopt these technologies more widely to reduce its dependence on foreign processing and strengthen its strategic position in the global mineral supply chain.

REGULATORY & POLICY CHALLENGES

1. Regulatory and permitting delays U.S. mine development faces long permitting timelines because of NEPA reviews, state-level mineral laws, and litigation. Many projects take a long time to reach production, reducing investor confidence and slowing domestic capacity.
2. Fragmented policy and limited midstream support: Federal incentives under the IRA and CHIPS Act center on manufacturing, while midstream processing and refining receive limited support. Several agencies share overlapping responsibilities without a unified national strategy, creating inconsistent guidance for companies. Many emerging technologies remain

in the “valley of death,” where they fail to progress from pilot-scale to commercial operations because of unclear regulatory pathways and limited long-term incentives.

3. **International and structural constraints:** Because the U.S. has not ratified the UN Convention on the Law of the Sea, it cannot obtain International Seabed Authority exploration rights, limiting participation in deep-sea mineral activities where China holds multiple claims. Some critical minerals, including gallium, germanium, and helium, have few or no economically viable domestic sources, which increases reliance on innovation and recycling. Yet U.S. recycling systems are fragmented, and technologies such as Joule-heating e-waste recycling and ARPA-E separation methods are still early-stage, leaving gaps in supply resilience.

RECOMMENDATION

1. Establish a National Critical Minerals Fast-Track Permitting Framework with Environmental Safeguards: The U.S. should create a clear and faster permitting process for critical mineral projects that use newer, lower-impact technologies like AI-enabled exploration, autonomous mining systems, and modular processing units. The framework should aim to complete approvals within 2 to 3 years while keeping strong environmental reviews and community consultation. Reforms can include expanding FAST-41²¹ to cover more critical mineral projects, setting fixed timelines for NEPA reviews, and improving coordination across federal agencies through dedicated mineral offices. Reducing the current 20–30 year timelines will make investment more attractive and help the U.S. move at the same pace as partners like Australia and Canada.

Rationale: A faster and more predictable permitting system will help the U.S. bring new projects online quickly, expand domestic supply, and reduce dependence on foreign processing.

2. Create a U.S. Critical Minerals Innovation and Commercialization Fund: The federal government should set up a dedicated fund to support innovation and commercialization across the critical mineral supply chain. The fund can follow the model of DARPA and ARPA-E by supporting research, pilot projects, and early commercial testing in areas such as AI-enabled exploration, autonomous mining, subsea robotics, advanced processing technologies, and recycling. Funding should focus on technologies that are already showing progress but need help moving from the lab into real industrial applications. Long-term, stable financing through demonstration plants, loan guarantees, and grants will help scale these innovations.

Rationale: Closing the commercialization gap will allow the U.S. to turn its scientific strengths into real industrial capacity and reduce reliance on China for key processing steps.

3. Expand Domestic Processing, Refining, and Recycling Capacity Through Targeted Incentives: The U.S. should increase midstream capacity by providing targeted incentives such as production tax credits, low-interest federal loans, and public-private processing hubs for minerals like rare earths, lithium, graphite, and battery materials. The government can also support investment by offering offtake agreements and price-stabilization tools. At the same time, a national recycling framework should be introduced to encourage the recovery of minerals from batteries, electronics, solar panels, and defense systems. Supporting recycling companies and motivating manufacturers to use recycled inputs will help create a circular supply chain.

Rationale: Improving processing and recycling capacity tackles the U.S.’s biggest supply-chain weakness and reduces the country’s heavy dependence on China’s refining sector.

4. Lead Internationally Through Strategic Alliances and Deep-Sea Governance: The U.S. should strengthen its cooperation with close partners like Australia, Canada, Japan, and the EU through joint research and shared processing facilities. To participate fully in global deep-sea mineral governance, the U.S. should also ratify the UN Convention on the Law of the Sea, which would allow it to take part in International Seabed Authority licensing and standards-setting. Stronger alliances and clearer rules will help diversify supply sources and reduce risks tied to concentrated processing in countries like China.

Rationale: Working with trusted partners and shaping global rules will improve supply security, reduce pressure on domestic sources, and give the U.S. more influence over emerging mineral markets.

5. Build Data and Workforce Foundations for an Innovation-Driven Supply Chain: The U.S. should develop a national critical minerals data system that provides real-time information on geology, production, processing, and recycling. Better data will help companies and policymakers make informed decisions and reduce exploration risks. At the same time, federal training programs, scholarships, and apprenticeships should support workforce development in geology, mining and processing engineering, refining, and robotics. These programs should link directly to new technologies like AI-enabled mining, advanced refining systems, and automation.

Rationale: Reliable data and a skilled workforce are essential for safely deploying new technologies, supporting regulatory oversight, and maintaining long-term competitiveness in the supply chain.

CONCLUSION

This paper shows that innovation is central to strengthening U.S. critical mineral supply chains and reducing dependence on China. While the United States has strong research capacity and emerging technologies in AI-enabled mining, deep-sea robotics, advanced processing, and recycling, these efforts remain fragmented and slow to scale. In contrast, China’s coordinated approach across mining, processing, and global investment continues to give it a strategic advantage. The main challenge for the U.S. is not the absence of resources or technology, but weak coordination, long permitting timelines, limited midstream capacity, and gaps in commercialization.

To remain competitive, the United States must move beyond pilot projects and adopt a coordinated national strategy that moves innovation across the entire supply chain. Faster permitting, targeted funding for commercialization, expanded processing and recycling capacity, stronger international partnerships, and investment in data and workforce development are all necessary steps. Without decisive action, U.S. reliance on foreign supply chains will persist. With the right policies, however, innovation can secure critical minerals, support clean energy and defense industries, and strengthen U.S. global competitiveness in the long term.²²

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